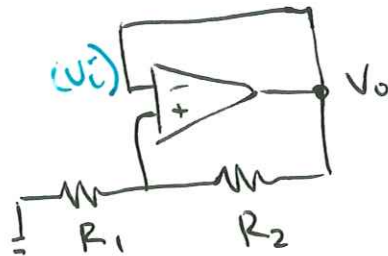
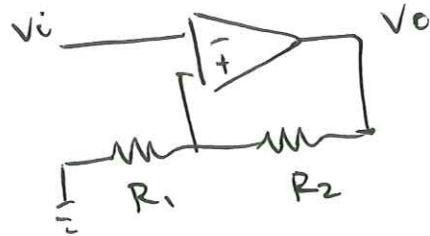


Oscillators



$$V_{TH+} = V_{OH} \frac{R_1}{R_1 + R_2} \quad V_{TH-} = V_{OL} \frac{R_1}{R_1 + R_2}$$

Inverting Schmitt Trigger

High input will cause toggle $\rightarrow V_o$ low (V_{OL})
 Low input will cause toggle $\rightarrow V_o$ high (V_{OH})

$\rightarrow$ Assuming α is ideal $f_b \rightarrow \infty$, $SR \rightarrow \infty$

V_o will toggle at $f_o = \infty$

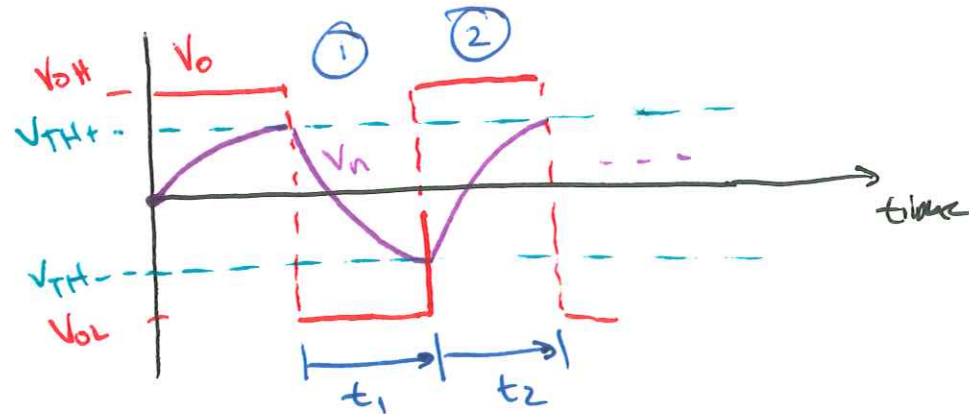
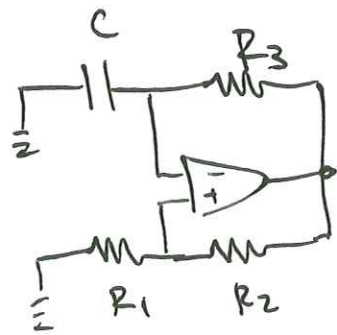
But! SR is limited (and so is f_b)

$\rightarrow V_o$ will toggle at some $f_o \neq \infty$

This will result in fast triangular wave!
 But f_o not controllable!

Can control f_o if delay between V_i & V_o ,
 that is a change in V_o takes time to propagate
 to V_i

Controlled f_o



Let's assume $f_o \ll f_{PB}, f_{\text{clk}}$

$$\textcircled{1} \quad V_n = V_{TH+} + (1 - e^{-t/\tau})(V_{OL} - V_{TH+})$$

t relative to last toggle

$$\textcircled{2} \quad V_n = V_{TH-} + (1 - e^{-t/\tau})(V_{OH} - V_{TH-})$$

If $V_{OH} = -V_{OL}$, then $V_{TH+} = -V_{TH-}$, rise time t_1 & t_2 equal

to find t_1 , take eq $\textcircled{1}$ and solve for $V_n = V_{TH-}$

$$t_1 = R_3 C \ln \left(\frac{V_{TH-} - V_{OH}}{V_{TH+} - V_{OH}} \right) \quad (3) = T/2$$

$$t_2 = R_3 C \ln \left(\frac{V_{OH} - V_{TH-}}{V_{OH} - V_{TH+}} \right) = T/2$$

$$f_o = \frac{1}{2 R_3 C \ln \left(\frac{V_{OH} - V_{TH-}}{V_{OH} - V_{TH+}} \right)} = \frac{1}{2 R_3 C \ln (1 + 2 R_1 / R_2)}$$

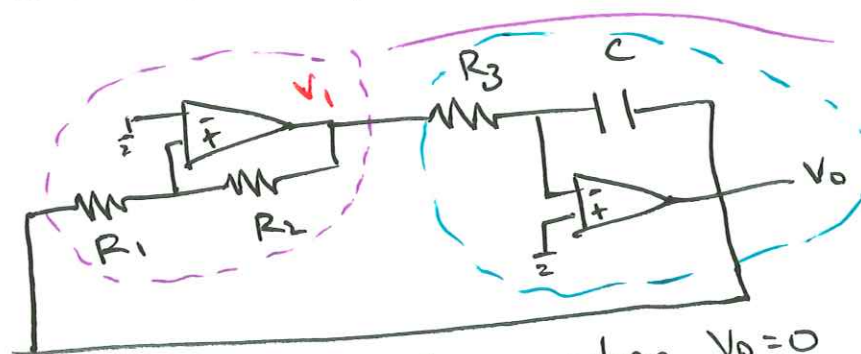
Square wave output!

Triangular Wave Generator

Can take previous circuit (or any other sq. wave output) and integrate!

Or... use integrator as V_n delay input.

Or... use integrator to control V_{TH+}/V_{TH-}



Non-inv. Schmitt Trigger

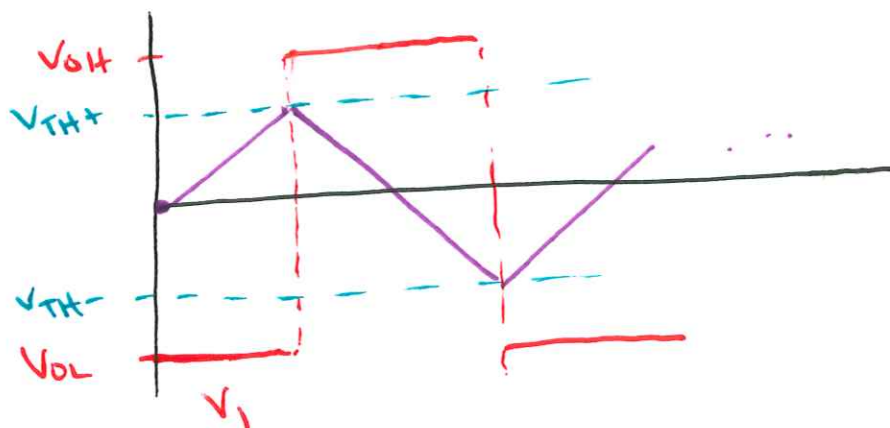
inv. integrator

find thresholds: Occur when $V_p = 0$

$$V_p = (V_1 - V_0) \frac{R_1}{R_1 + R_2} + V_0 = 0$$

V_1 will be either V_{OH} or V_{OL}

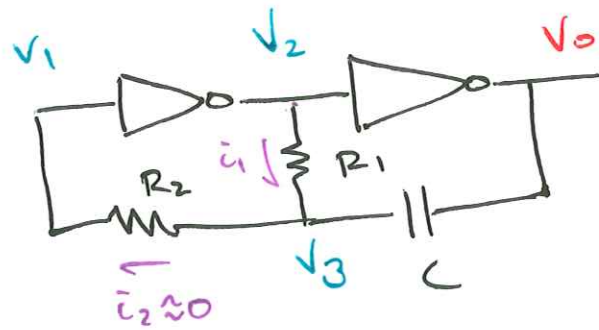
$$V_{TH+/-} = \frac{R_1}{R_2} V_{OH/OL}$$



$$f_0 = \frac{R_2 / R_1}{4R_3 C}$$

DC
 → Bias of integrator forced to 0!

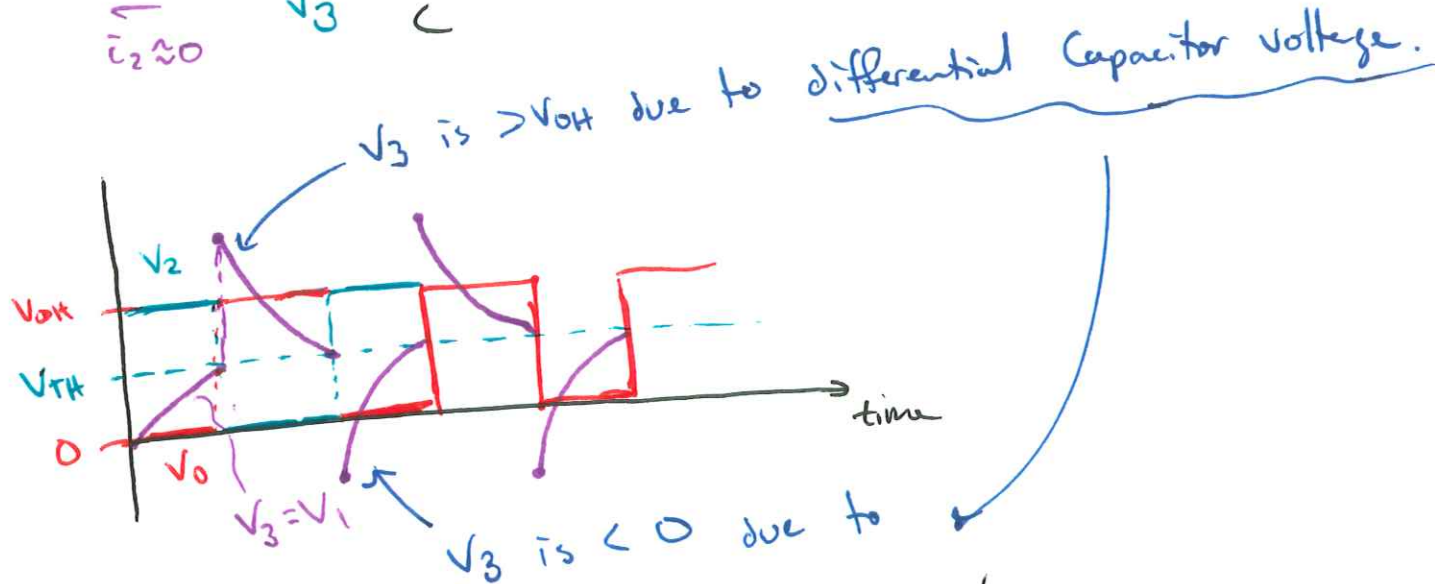
Another Square Wave Gen.



Digital inverter

$$V_i < V_{TH} \rightarrow V_o = V_{OH} (\approx V_{supply})$$

$$V_i > V_{TH} \rightarrow V_o \approx 0 (V_{OL})$$

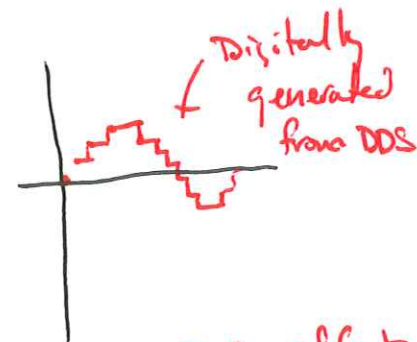


$$f_0 = 1 / (R_1 C \ln 3) = 0.91 / (R_1 C)$$

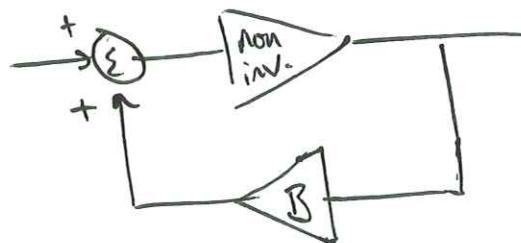
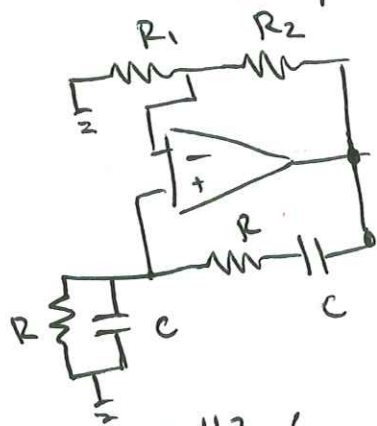
Sine wave Generation

1 way: Square OR ~~tri~~ wave generator with sharp LPF with corner just above f_0 LPF design and manufacture Difficult!

2nd: DDS devices Direct Digital Synth.
effectively a DAC with freq. control.



3rd: Wien-Bridge.



Assume f_0 of non-inv. amp $\gg$ RC effects.

$$B(j\omega) = \frac{R \parallel Z_C}{R \parallel Z_C + R + Z_C} \quad Z_C = 1/j\omega C$$

$$B(j\omega) = \frac{1}{3} + j2\pi fRC + \frac{1}{j2\pi fRC}$$

$$= \frac{1}{3} + j\left(\frac{f}{f_0} - \frac{f_0}{f}\right)$$

$$f_0 = 1/2\pi RC$$

Resonance will occur when $B(jf)$ max.
 $\rightarrow f/f_0 = f_0/f \rightarrow f_0 = f \quad (f_r = f_0)$



$$T_p(jf) = A_{non-inv} B(jf) = \frac{1 + \frac{R_2}{R_1}}{3 + j(f/f_0 - f_0/f)}$$

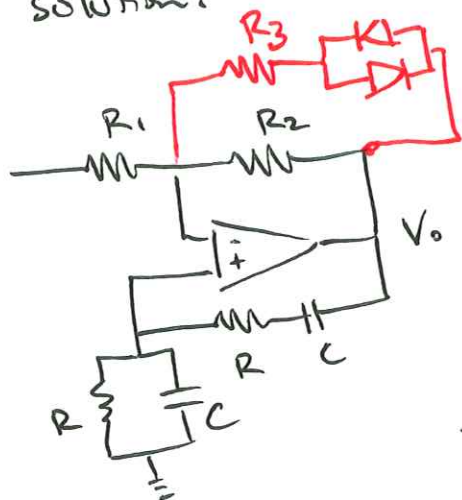
$$T_p(jf_0) = \frac{1 + \frac{R_2}{R_1}}{3} \rightarrow \text{want this exactly at } 1!$$

why? this is effectively the -180° point ($f_0 = f_{-180}$)
 since positive feedback, $\nexists T_p(jf) \neq 0$ when f_{-180}

$$1 = \frac{1 + \frac{R_2}{R_1}}{3} \rightarrow \frac{R_2}{R_1} = 2 \rightarrow \text{sin wave at } f_0 \text{ (ideally)}$$

Very Difficult, near impossible to have $T_p(jf_0) = 1$ due to tolerances...

→ solution:



→ if Voltage between V_0 & V_n large, then
 $T_p(jf_0) \approx \frac{1 + \frac{R_2 \parallel R_3}{R_1}}{3} \rightarrow \text{Reduces magnitude of } T_p$

if voltage low, diodes off:

$$T_p(jf_0) \approx \frac{1 + \frac{R_2}{R_1}}{3}$$

Design: Select R_2 such that $T_p(jf_0)$ just larger than 1
 R_3 such that $T_p(jf_0)$ just smaller than 1
 ← Diodes off
 ← Diodes on.